Simplified Quant Trading System

Final Project Report

1. Project Overview

This project builds a simplified quantitative trading workflow: process OHLCV data, calculate indicators, generate trading signals, simulate market and limit order execution, and evaluate performance against a buy-and-hold baseline.

2. Dataset Description

The submitted dataset is a reproducible synthetic OHLCV series with 756 business-day rows from 2022-02-08 to 2024-12-31. Synthetic data keeps the project runnable without network access while preserving realistic open, high, low, close, and volume structure.

3. Indicators Used

The workflow calculates simple returns, cumulative returns, 20-day and 50-day moving averages, 20-day rolling volatility, and 20-day average volume. Moving averages smooth price trends, volatility estimates changing risk, and volume average highlights market activity.

4. Trading Strategy Logic

The main strategy is a Moving Average Crossover: BUY when the 20-day moving average is above the 50-day moving average, and SELL when it is not. A Mean Reversion strategy is also included: BUY when price is materially below its 20-day rolling mean and SELL when price is materially above it.

5. Execution Logic

Market orders use a fixed $0.10 spread: bid equals close minus half the spread, and ask equals close plus half the spread. BUY orders fill at ask and SELL orders fill at bid. Limit BUY orders fill only when ask is less than or equal to the limit price; limit SELL orders fill only when bid is greater than or equal to the limit price.

To avoid lookahead bias, signals are shifted by one bar before execution. A signal observed on day t can only affect trading on day t+1.

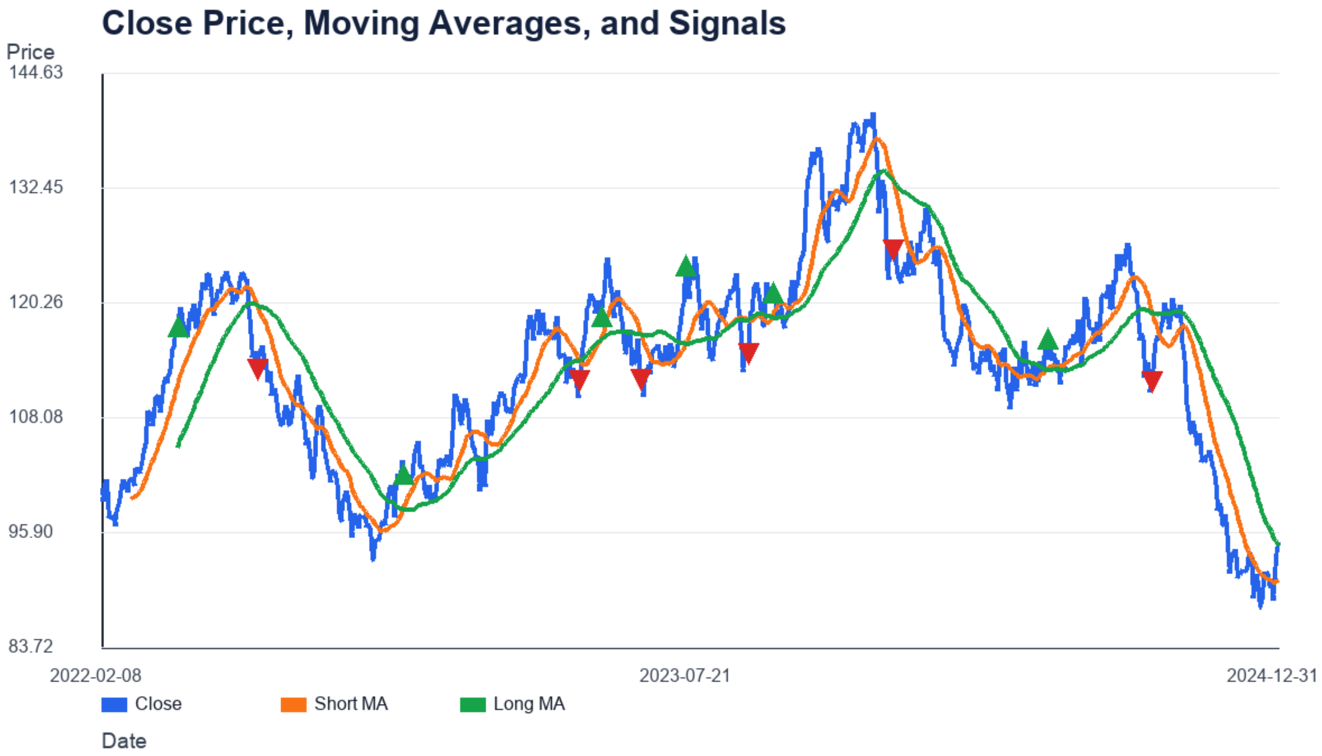
Limit order demonstration: a BUY limit at $86.97 produced 0 fills, while a BUY limit at $98.96 produced 1 fill.

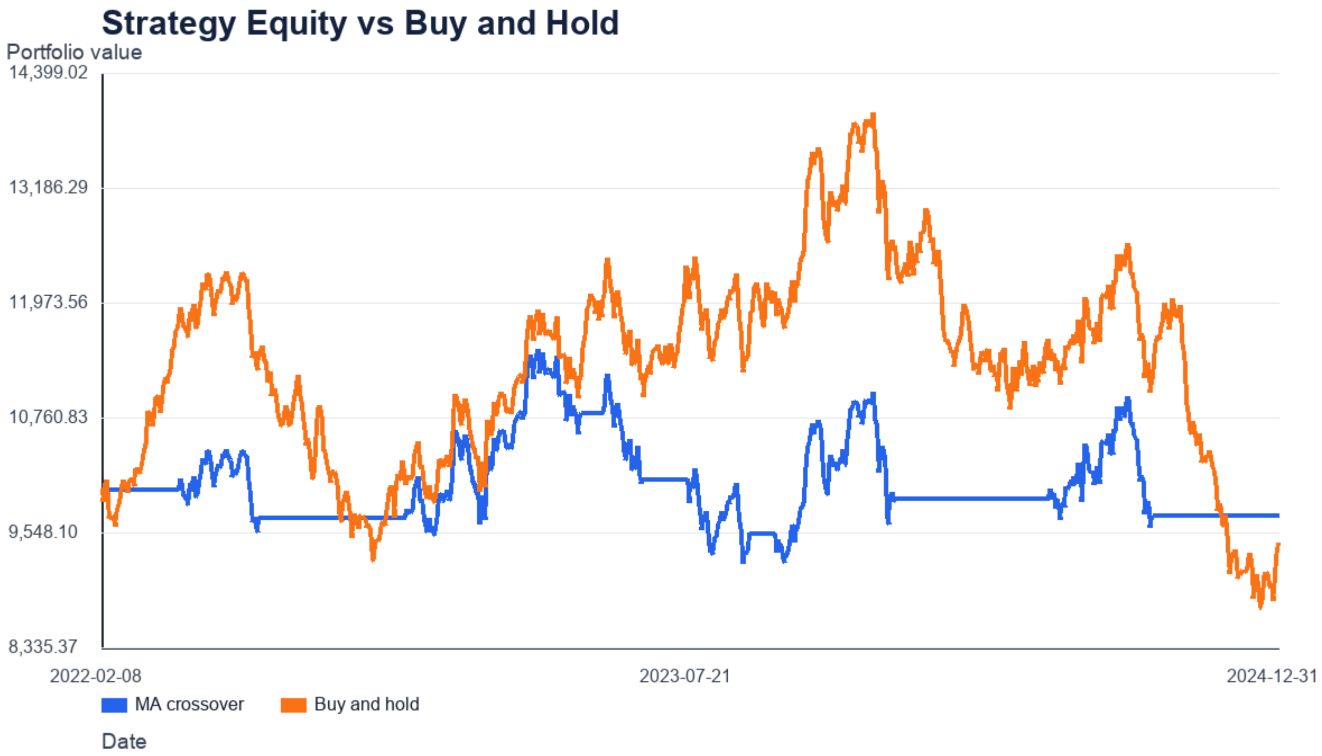
6. Results and Performance

Strategy	Final Equity	Return	Trades	Wins	Losses	Sharpe	Max DD
MA Crossover	$9,738.14	-2.62%	12	2	4	0.01	-19.70%
Mean Reversion	$10,434.90	4.35%	28	9	5	0.17	-27.99%
Buy and Hold	$9,432.22	-5.68%	1	-	-	-	-37.39%

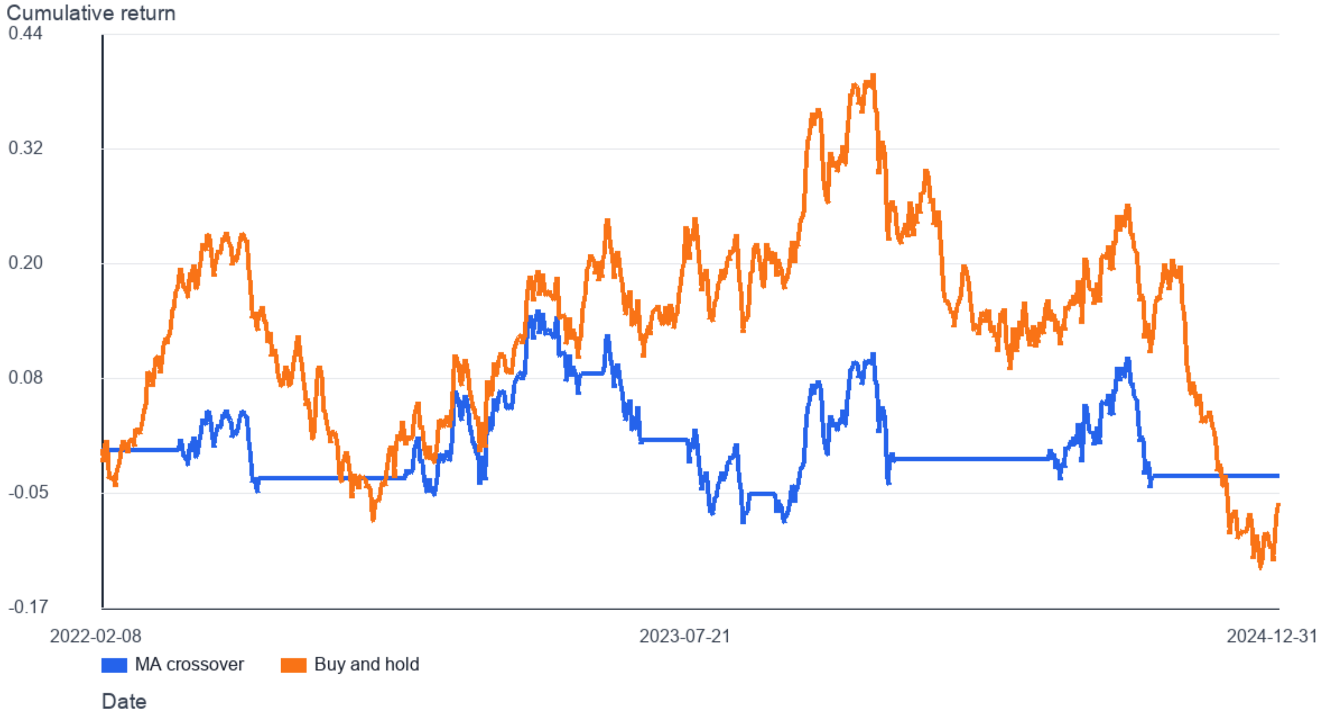
Mean Reversion performed best on this sample, while Moving Average Crossover finished ahead of Buy and Hold by avoiding some weaker periods. The passive baseline stayed fully invested and carried the deepest drawdown.

7. Charts and Visualizations

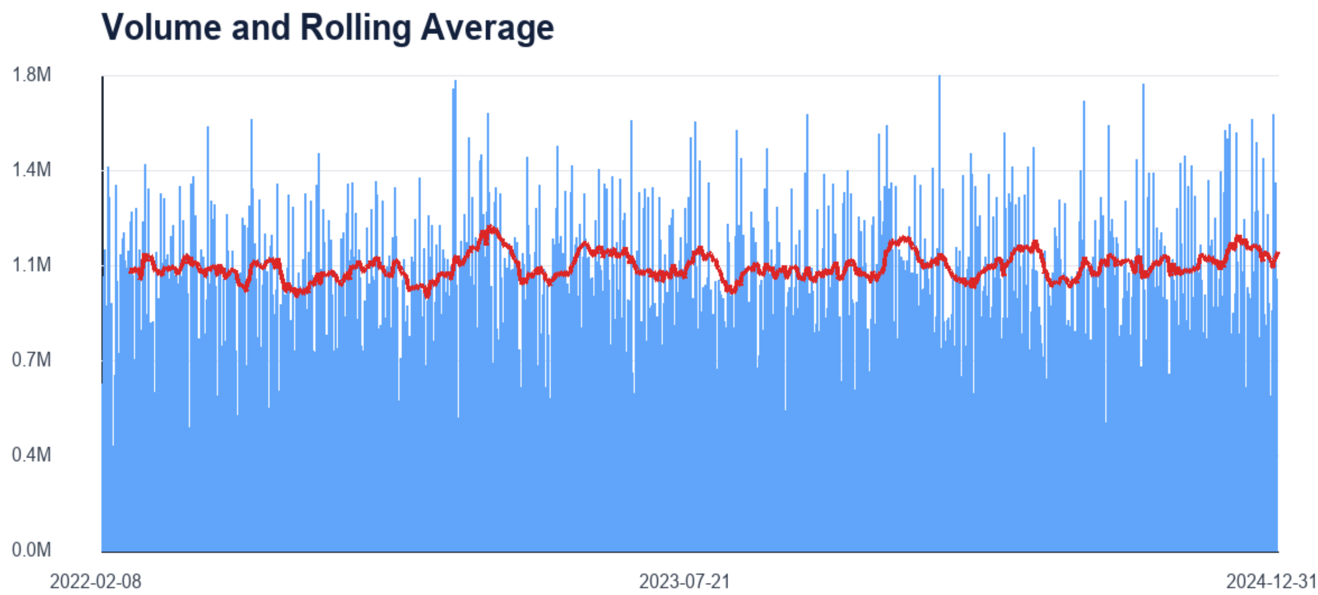


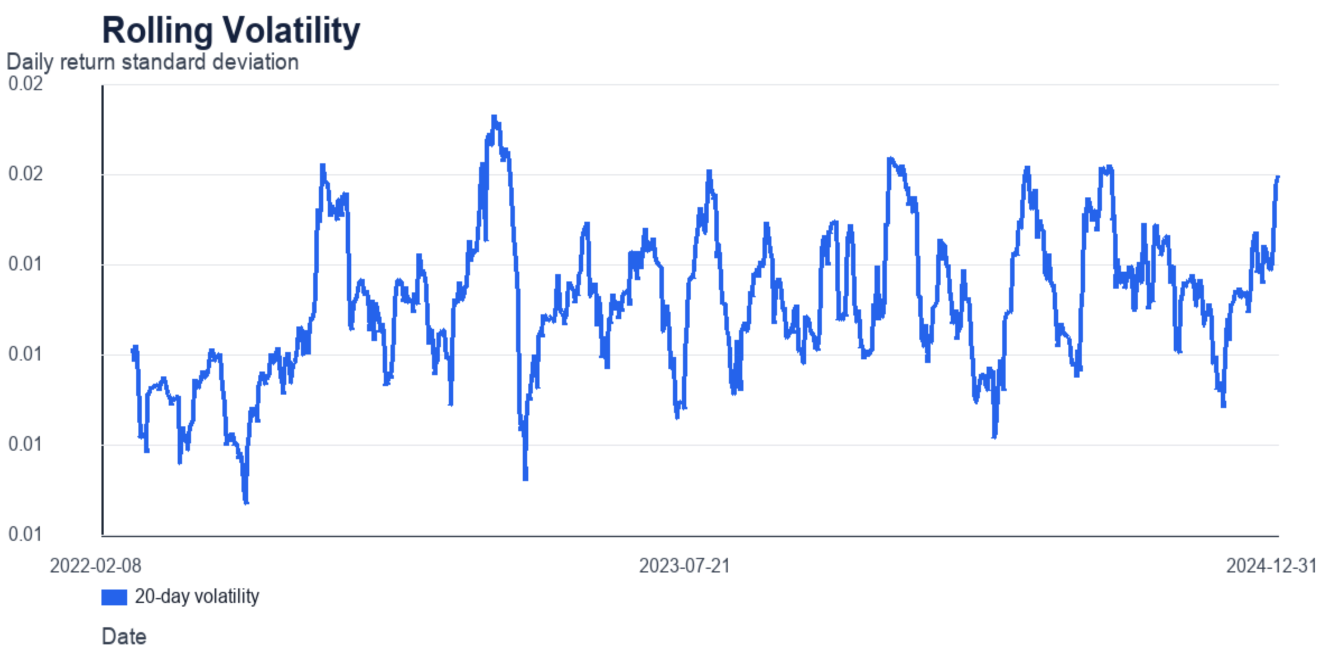


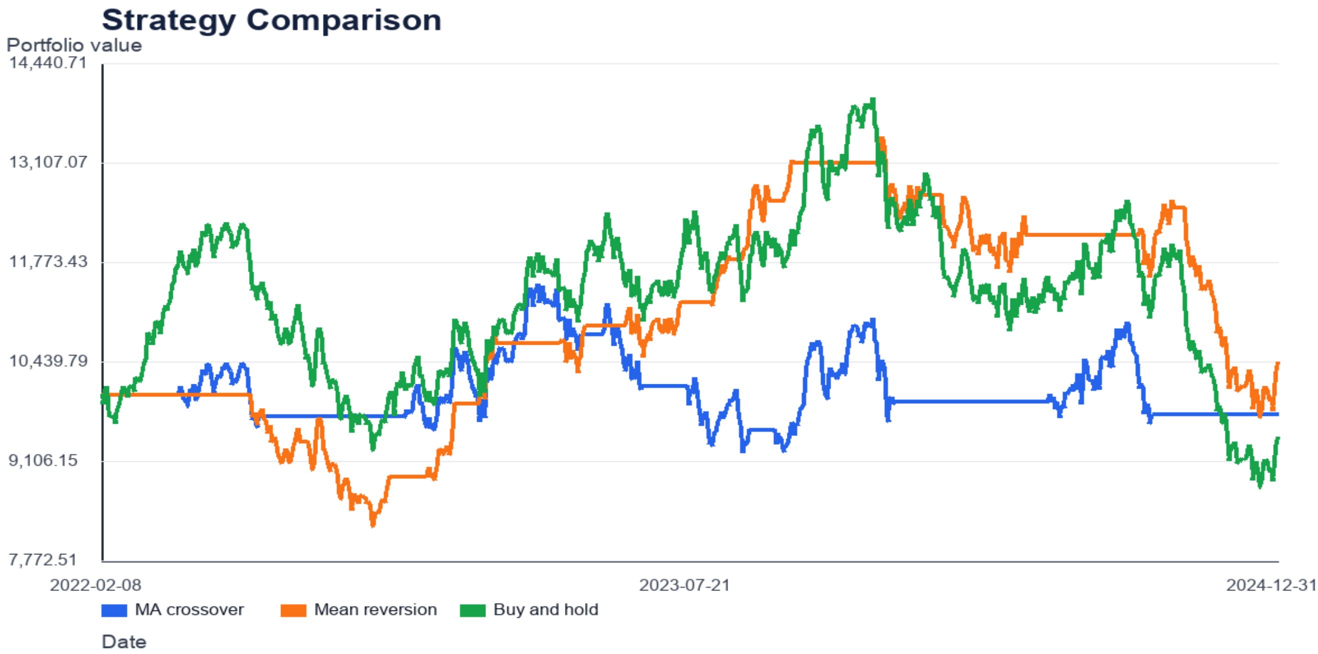
Cumulative Returns



Additional Market Diagnostics







8. Key Observations

The crossover strategy works best when trends persist after the short average crosses the long average. In choppy regions, crossovers can arrive late and create whipsaw trades. The mean reversion rule reacts differently by buying weakness, which can help in sideways markets but can struggle during persistent downtrends.

9. Limitations

The system uses one synthetic asset, fixed spread, fractional shares, and in-sample parameter choices. It does not model market impact, taxes, borrow costs, order queue priority, or changing liquidity. Results should be treated as a research exercise rather than a live trading recommendation.

10. Conclusion

The project demonstrates the full quantitative trading pipeline from data preparation to signal logic, execution, and evaluation. The most important correctness control is the one-bar execution delay, which prevents the strategy from using future information.